

Ferrari 20 More Rounds

Coset Union Upper-Bound Lemma and Certified Bounded Application

Verdict

PROMOTE_MATH_VALUE_HIGH_AS_FINITE_QUOTIENT_LEMMA_NOT_GLOBAL_BREAKTHROUGH

Active Binary

Can the coset-size upper-bound obstruction be promoted from bounded evidence into a valuable finite quotient lemma and verified branch certificate?

Result

The result is now worth something as a finite quotient lemma. It is not just branch enumeration. Each danger modulus cuts out at most one coset in the target subfiber quotient, and a strict union-size upper bound proves full coverage impossible for the certified branches.

Lemma

Let X be a finite subfiber quotient. For each danger modulus p , suppose the covered set inside X is either empty or one coset S_p of a subgroup kernel. If $\sum_p |S_p| < |X|$, then $\bigcup_p S_p$ is not X . Therefore the branch cannot cover the full target subfiber.

Proof

The union cardinality is bounded by the sum of component cardinalities: $|\bigcup_p S_p| \leq \sum_p |S_p|$. If $\sum_p |S_p| < |X|$, then $|\bigcup_p S_p| < |X|$, so the union cannot equal X .

Application Certificate

Metric	Value
Branch rows checked	212
Branches	106
All rows prove noncover	True
Worst coverage ratio	0.125
Min surviving lower bound	$\{ "(20,20)": 6, "(40,40)": 6 \}$

Twenty Rounds

Round 1 - PROMOTE_LOCK

Active binary locked to coset upper-bound promotion, not new framework.

Round 2 - PROMOTE_FULL_RACE8_BRANCH_SET

Imported 106 Race8 q3 branches: 52 replayed plus 54 previously skipped.

Round 3 - PROMOTE_EXACT_SUBFIBERS

Defined target subfibers for (20,20) and (40,40) using $k=k_0+120a$, $l=l_0+240b$.

Round 4 - PROMOTE_COSET_EQUATION

Derived the single-modulus equation $(2^{120})^a(3^{240})^b = -(2^{k_0} 3^{l_0})^{(-1)} \bmod p$.

Round 5 - PROMOTE_COSET_STRUCTURE

Promoted each modulus hit set to empty or one coset of a kernel in a finite quotient.

Round 6 - PROMOTE_LEMMA

Proved the union upper-bound lemma: if sum coset sizes < quotient size, full coverage is impossible.

Round 7 - PROMOTE_REPLAYED_BRANCHES

Applied the lemma to all replayed branches. 52/52 passed.

Round 8 - PROMOTE_SKIPPED_BRANCH_REPAIR

Applied the same lemma to previously skipped branches. 54/54 passed.

Round 9 - PROMOTE_BOTH_TARGETS

Verified both target cells separately. Neither has a full-cover branch.

Round 10 - PROMOTE_MARGIN

Worst-case coverage ratio was 0.125, still far below full coverage.

Round 11 - PROMOTE_SURVIVAL_MARGIN

Minimum surviving lower bound is 6 for each target cell.

Round 12 - PROMOTE_CLAIM_SEPARATION

Separated theorem from application: lemma is general; q_3 range application is bounded.

Round 13 - PROMOTE_DISCIPLINE

Rejected global overclaim: no statement about q_3 outside 301..997 or unbounded depth.

Round 14 - PROMOTE_REFeree_VALUE

Referee value identified: converts skipped brute-force cases into a finite quotient certificate.

Round 15 - PROMOTE_FORMALIZATION_TARGET

Formalization target extracted: finite quotient, cosets, and union-cardinality inequality.

Round 16 - PROMOTE_REPLAY_VERIFIER

Verifier script created to check all row inequalities and summary values.

Round 17 - PROMOTE_MATH_VALUE_HIGH_LEMMA

Math value upgraded from bounded enumeration to finite quotient lemma plus certified application.

Round 18 - PROMOTE_NO_OVERCLAIM

Breakthrough value remains not promoted; global theorem not claimed.

Round 19 - PROMOTE_NEXT_BINARY

Next binary selected only after verdict: can this inequality be proven for all actual danger-closure branches from the parent family?

Round 20 - PROMOTE_FINAL

Final verdict: high-value lemma target and certified bounded application.

What This Proves

- general finite quotient lemma
- bounded branch noncoverage certificate for all 106 Race8 branches including previously skipped branches
- noncoverage without full fiber enumeration

What This Does Not Prove

- q3 outside 301..997
- unbounded recursive depth
- arbitrary non-danger moduli
- global Erdos problem #203

Math Value

High as a finite quotient lemma and certified method. Not a global breakthrough.

Exact Next Binary

Can the coset-size inequality be proven for every actual danger-closure branch generated from the Race8 parent family, rather than only q3 in 301..997?